Ecuații diferențiale liniare de ordin superior

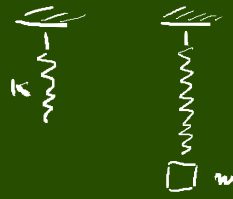
ec. omogene $x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x = 0$

$\lambda^n + a_1 \lambda^{n-1} + \dots + a_n = 0$

$x_0(t) = C_1 x_1(t) + \dots + C_n x_n(t)$ $\quad C_1 \dots C_n \in \mathbb{R}$

$x_1(t), \dots, x_n(t)$ sist. fundam. de sol.

exemplu.



$x(t)$

$x'' + \underbrace{\frac{\mu}{m}}_{2a} x' + \underbrace{\frac{k}{m}}_{\omega^2} = 0$

$x'' + 2a x' + \omega^2 = 0$

$\lambda^2 + 2a\lambda + \omega^2 = 0$

$\Delta = 4(a^2 - \omega^2)$

k - d. de elasticitate
m masa
μ coef. de frecare
$x(0) = x_0$
$x'(0) = v_0$

I $\quad \Delta < 0 \qquad 0 \le a < \omega$

I.1 $\quad a = 0$ (fără frecare)

$\Delta = -4\omega^2 \qquad \lambda_{1,2} = \pm i\omega \Rightarrow x(t) = C_1 \cos \omega t + C_2 \sin \omega t$

frecvență $\frac{2\pi}{\omega}$

amplitudine $\sqrt{C_1^2 + C_2^2}$

$\varphi = \operatorname{arctg} \frac{C_1}{C_2}$

I.2 $a \neq \omega$ $\quad 0 < a < \omega$ (facem singuri)

$\Delta = 4(a^2 - \omega^2) = -4\nu^2 \qquad \nu = \sqrt{\omega^2 - a^2} < \omega$

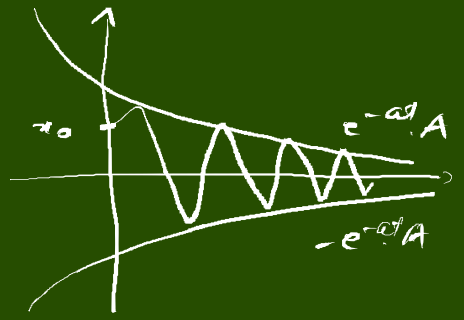
$\Rightarrow \lambda_{1,2} = \frac{-2a \pm 2\nu i}{2} = -a \pm i\nu$

$\Rightarrow x(t) = C_1 e^{-at} \cos \nu t + C_2 e^{-at} \sin \nu t$

$= \underbrace{e^{-at}}_{\to 0} \underbrace{(C_1 \cos \nu t + C_2 \sin \nu t)}_{\text{mărginită}} \xrightarrow{t \to \infty} 0$

(→ perioada $\frac{2\pi}{\nu}$

frecvență $\frac{\nu}{2\pi} < \frac{\omega}{2\pi}$



$x(0) = x_0$

$x'(0) = v_0$

$\Rightarrow C_1 = \dots$

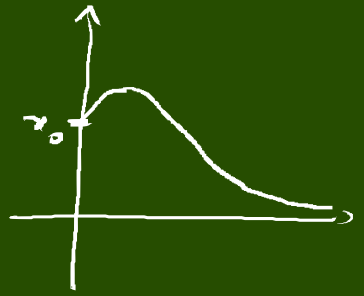
$C_2 = \dots$

II $\quad \Delta = 0 \quad$ (pod. kritické) $\qquad a = \omega$

$\lambda^2 + 2a\lambda + \omega^2 = 0$ $\qquad \left(\frac{\mu}{2m} = \frac{k}{m} \Rightarrow \mu = 2k \right)$

$\lambda_1 = \lambda_2 = -a$

$\Rightarrow x(t) = C_1 e^{-at} + C_2 e^{-at} \cdot t \xrightarrow{t \to \infty} 0$



III $\Delta > 0$ $\quad a > \omega$ $\quad \Delta = 4(a^2 - \omega^2) = 4\sigma^2$

$\sigma = \sqrt{a^2 - \omega^2} < a$

$$\lambda_{1,2} = \frac{-2a \pm 2\sigma}{2} = \begin{cases} -a - \sigma < 0 \\ -a + \sigma < 0 \end{cases}$$

$$\Rightarrow x(t) = C_1 e^{(-a-\sigma)t} + C_2 e^{(-a+\sigma)t} \xrightarrow{t \to \infty} 0$$

$$= e^{-at}\left(C_1 e^{-\sigma t} + C_2 e^{\sigma t}\right)$$

$A \sin(\omega t + \varphi_0)$

$A \operatorname{sh}(\sigma t + \varphi_0)$ $\quad A = ?$ (temă) $\quad \varphi_0 = ?$

Ecuații diferențiale liniare neomogene de ordin superior

$$x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x = b(t)$$

$x(t) = x_0(t) + x_p(t)$

sol. ec. dif. omog. $\quad$ sol. part. a ec. dif. neomogene

Metoda variației constantelor

$x_0(t) = C_1 x_1(t) + \dots + C_n x_n(t)$ $\quad C_1, \dots C_n \in \mathbb{R}$

$x(t) = D_1(t) x_1(t) + \dots + D_n(t) x_n(t)$ unde funcțiile $D_1(t), \dots, D_n(t)$

Verificăm ecuația:

$$\begin{cases} D_1'(t) x_1(t) + \dots + D_n'(t) x_n(t) = 0 \\ \vdots \\ D_1'(t) x_1^{(n-2)}(t) + \dots + D_n'(t) x_n^{(n-2)}(t) = 0 \\ D_1'(t) x_1^{(n-1)}(t) + \dots + D_n'(t) x_n^{(n-1)}(t) = b(t) \end{cases}$$

Exemplu $\quad x'' + x = \operatorname{tg} t \quad$ $x(0) = 0, \; x'(0) = 0, \; t \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$x'' + x = 0$

$\lambda^2 + 1 = 0 \Rightarrow \lambda_{1,2} = \pm i \Rightarrow x_1(t) = \cos t$

$x_2(t) = \sin t$

$\Rightarrow x_0(t) = C_1 \cos t + C_2 \sin t \quad C_1, C_2 \in \mathbb{R}.$

$$x'' + x = \operatorname{tg} t$$

$$x(t) = D_1(t)\cos t + D_2(t)\sin t$$

$$\begin{cases} D_1'(t)\cos t + D_2'(t)\sin t = 0 \\ -D_1'(t)\sin t + D_2'(t)\cos t = \operatorname{tg} t \end{cases}$$

$$\begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = 1$$

$$D_1'(t) = \begin{vmatrix} 0 & \sin t \\ \operatorname{tg} t & \cos t \end{vmatrix} = \frac{\sin^2 t}{\cos t}$$

$$D_2'(t) = \begin{vmatrix} \cos t & 0 \\ -\sin t & \operatorname{tg} t \end{vmatrix} = \sin t \Rightarrow D_2(t) = -\cos t + k_2$$

$$D_1(t) = \int \frac{\sin^2 t}{\cos t}\,dt = \int \left(\frac{1}{\cos t} - \cos t\right)dt = \int \frac{1}{\cos t}\,dt - \sin t$$

$\sin t = u$, $\cos t\,dt = du$

$\operatorname{tg}\frac{t}{2} = v$

$$\int \frac{u^2-1+1}{1-u^2}\,du = -u - \int \frac{1}{u^2-1}\,du =$$

$$= -u - \frac{1}{2}\int\left(\frac{1}{u-1} - \frac{1}{u+1}\right)du$$

$$= -u - \frac{1}{2}\ln\left|\frac{u-1}{u+1}\right|$$

$$D_1(t) = -\sin t - \frac{1}{2}\ln\left|\frac{\sin t - 1}{\sin t + 1}\right| + k_1$$

$$= -\sin t - \frac{1}{2}\ln\frac{1-\sin t}{1+\sin t} + k_1$$

$$\Rightarrow \quad x(t) = D_1(t)\cos t + D_2(t)\sin t$$

$$= \left(-\sin t - \frac{1}{2}\ln\left(\frac{1-\sin t}{1+\sin t}\right) + k_1\right)\cos t + \left(-\cos t + k_2\right)\sin t$$

$$= \overbrace{-\sin 2t - \frac{1}{2}\cos t \cdot \ln\left(\frac{1-\sin t}{1+\sin t}\right)}^{x_p(t)} + \overbrace{k_1\cos t + k_2\sin t}^{x_0(t)}$$

$x(0) = 0$
$x'(0) = 0$ $\Rightarrow$ $k_1 = \dots$ $k_2 = \dots$

$x^{(n)} + a_1 x^{(n-1)} + \dots + a_n x = b(t)$
$\lambda^n + a_1 \lambda^{n-1} + \dots + a_n = 0$

$x(t) = x_0(t) + x_p(t)$

Metoda coeficienților nedeterminați

I $b(t) = P(t) e^{\lambda t}$ $\quad P \in \mathbb{R}[x]$, $\lambda \in \mathbb{R}$

$$x_p(t) = \begin{cases} Q(t) e^{\lambda t}, & Q \in \mathbb{R}[x],\ \operatorname{grad} Q = \operatorname{grad} P \text{ dacă } \lambda \text{ nu e} \\ & \text{rădăcină a ecuației caracteristice} \\ t^k Q(t) e^{\lambda t}, & \text{dacă } \lambda \text{ e rădăcină cu multiplicitatea } k \\ & \text{a ecuației caracteristice} \end{cases}$$

($\lambda = 0 \Rightarrow b(t) = P(t)$)

II $b(t) = e^{\alpha t}\left(P(t)\cos\beta t + Q(t)\sin\beta t\right)$ $\quad P, Q \in \mathbb{R}[x]$, $\alpha, \beta \in \mathbb{R}$

$$x_p(t) = \begin{cases} e^{\alpha t}\left(P_1(t)\cos\beta t + Q_1(t)\sin\beta t\right), & P_1, Q_1 \in \mathbb{R}[x] \\ & \operatorname{grad} P_1 = \operatorname{grad} Q_1 = \max(\operatorname{grad} P, \operatorname{grad} Q) \\ & \text{dacă } \lambda = \alpha + i\beta \text{ nu e răd. a ec. caracteristice} \\ t^k e^{\alpha t}\left(P_1(t)\cos\beta t + Q_1(t)\sin\beta t\right), & P_1, Q_1 \end{cases}$$

dacă $\lambda = \alpha + i\beta$ este rădăcină a ec. caract. cu multipl. k

exemple:

1) $x'' + 4x = 5$

$x'' + 4x = 0$

$\lambda^2 + 4 = 5$ $\quad\Rightarrow\quad \lambda^2 = 1 \quad \lambda_{1,2} = \pm 1$

$\lambda^2 + 4\lambda = 0$ $\quad\Rightarrow\quad \lambda_1 = 0, \ \lambda_2 = -4$

$\lambda^2 + 4 = 0$ $\quad\Rightarrow\quad \lambda_{1,2} = \pm 2i$

$x_0(t) = C_1 \cos 2t + C_2 \sin 2t$

$x_p(t) = ?$ $\qquad b(t) = 5 = P(t) e^{\lambda t}$

$\lambda = 0$ $\qquad P(t) = 5$

rădăcină? NU

$\Rightarrow x_p(t) = Q(t) e^{\lambda t} = a \qquad a \in \mathbb{R}$

$x_p'(t) = 0, \quad x_p''(t) = 0$

$0 + 4a = 5 \qquad a = 5/4$

$x_p(t) = 5/4$

$x(t) = x_0(t) + x_p(t) = C_1 \cos 2t + C_2 \sin 2t + \frac{5}{4}$

2) $x'' - 4x' + 5x = e^{2t} \cos t$

$x'' - 4x' + 5x = 0$

$\lambda^2 - 4\lambda + 5 = 0 \quad (\lambda - 2)^2 + 1 = 0 \quad 2 \pm i$

$x_0(t) = C_1 e^{2t} \cos t + C_2 e^{2t} \sin t$

$b(t) = e^{2t} \cos t = e^{\alpha t}(P(t) \cos \beta t + Q(t) \sin \beta t)$

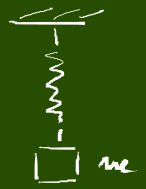
$\alpha = 2 \quad \beta = 1 \quad P(t) = 1 \quad Q(t) = 0$

$\lambda = \alpha + i\beta = 2 + i$ rădăcină? DA $k = 1$

$\Rightarrow x_p(t) = t^1 e^{2t} (\underbrace{P_1(t)}_{a} \cos t + \underbrace{Q_1(t)}_{b} \sin t)$

$a = ? \quad b = ?$ temă $\leftarrow a, b = ?$ metoda variației constantelor

3)



$F_{ext}(t) = m(1+\cos t)$

$m = 10\ kg \qquad k = 10\ N/m$

$h = 0$

$$x'' + \frac{k}{m}x = \frac{m}{m}(1+\cos t)$$

$$\begin{cases} x'' + x = 1 + \cos t \\ x(0) = 0 \\ x'(0) = 0 \end{cases}$$

$\Rightarrow$

$x'' + x = 0$

$\lambda^2 + 1 = 0 \qquad \pm i$

$x(t) = C_1 \cos t + C_2 \sin t$

$x'' + x = 1 + \cos t$

$x_{p_1} \quad x'' + x = 1 \quad \Rightarrow x_{p_1}(t) = 1$

$x_{p_2} \quad x'' + x = \cos t.$

$x_p(t) = x_{p_1}(t) + x_{p_2}(t)$ soluție pt $x'' + x = 1 + \cos t$

$x'' + x = \cos t$

$b(t) = \cos t = e^{\alpha t}(P(t)\cos\beta t + Q(t)\sin\beta t)$

$\alpha = 0 \qquad \beta = 1 \qquad P(t) = 1 \qquad Q(t) = 0$

$\lambda = \alpha + i\beta = i$ ← rădăcină cu multiplicitate $k = 1$

$\rightarrow x_p(t) = t(a\cos t + b\sin t)$

$x_p'(t) = a\cos t + b\sin t + t(-a\sin t + b\cos t)$

$x_p''(t) = -2a\sin t + 2b\cos t + t(-a\cos t - b\sin t)$

$-2a\sin t + 2b\cos t + t(-\underline{a\cos t} - \underline{\underline{b\sin t}}) + t(\underline{a\cos t} + \underline{\underline{b\sin t}}) = \cos t$

$\sin t : \quad -2a = 0 \quad \Rightarrow \quad a = 0$

$\cos t : \quad 2b = 1 \quad \Rightarrow \quad b = \frac{1}{2}$

$x_{p_2}(t) = \frac{1}{2} t\sin t$ $\qquad \frac{1}{i}\sin t + \frac{1}{2} t\cos t$

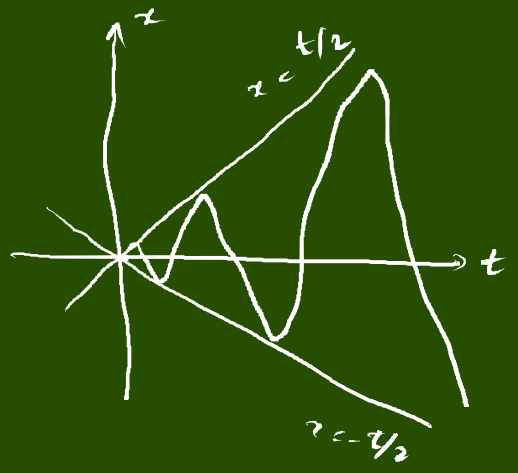
$\Rightarrow x(t) = \underbrace{C_1\cos t + C_2\sin t}_{x_0(t)} + \underbrace{1}_{x_{p_1}(t)} + \underbrace{\frac{1}{2} t\sin t}_{x_{p_2}(t)}$

$x(0) = 0 \Rightarrow C_1 + 1 = 0 \quad C_1 = -1$

$x'(0) = 0 \Rightarrow C_2 = 0$

$\Rightarrow x(t) = \underbrace{-\cos t}_{A=1,\ 2\pi} + \underbrace{1}_{C} + \underbrace{\frac{1}{2}t \sin t}$

Recomandă



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exercițiu.

31 ian
- 9 11:00 — 2 gr. — A 02 — 1 ora
- 9 14:00 — 2 gr. — A 02 — 1 ora

fizică

1) final 2 ore / 5 exerciții
mai (parțial) → max. 1 h
demonstrație + acasă.

2) ——— h ———

——— y ———

- - - - - -

$1{,}5 \le x < 2$ — 1 ex. 15'.
$1 \le x < 1{,}5$ — 2 ex. 30'.
$0{,}5 \le x < 1$ — 3 ex. 45'.
$0 \le x < 0{,}5$ — 4 ex. 1 h

Mercuc 8^{00} 31.ian 8^{00} – 10^{00} final.
10^{00} – 10^{15} încărcare
10^{30} – 11^{30} oral (partial) 15' pauză
max 1 h

Examenul final – materia

I – Pseudosoluția unui sistem incompatibil; regresia liniară
II – Matrice simetrice; forme pătratice ⟨ optimizare conice
III – Descompunerea valorilor singulare.
IV – Ecuații diferențiale de ordinul I liniar
V – Sisteme de ec. dif. liniare; portret de fază
VI – Ecuații dif. lin. de ordin superior

Consultație V 14^{00} – ~~16^{00}~~ 15^{00}
sau ~~S 9^{00} –~~ sau (D 9^{00} – h^{00})